Trust-Gated Capability Control: Breaking the Trust–Vulnerability Paradox in Multi-Agent LLM Systems

***Abstract*—Layered trust models for multi-agent large language model systems remain largely conceptual: they describe which dimensions of trust matter but do not specify how trust at different layers combines, how layer importance is set at runtime, or how trust should govern what an agent is allowed to do. This gap is consequential because increasing inter-agent trust is known to raise task success while simultaneously enlarging exposure to exploitation, a tension recently formalized as the Trust-Vulnerability Paradox. We make a five-layer trust stack operational through three contributions. First, we define a cross-layer synergy operator that propagates prerequisite-layer deficits into dependent layers, and a generalized-mean composite trust that recovers the weakest-link rule as a limiting case, with provable bounds. Second, we ground per-layer importance weights in observed failures through a no-regret online estimator that tracks which layer is currently most responsible for harm. Third, we introduce Trust-Gated Capability Control, a runtime mechanism that issues short-lived, revocable capability grants only when cross-layer composite trust and the relevant prerequisite layers clear capability-specific thresholds. We prove that this mechanism breaks the paradox: a stealthy compromise that inflates behavioral trust while degrading a prerequisite layer cannot escalate privilege, and we give a closed-form, provably conservative trust fixed point with a bounded-latency revocation guarantee. A numerical study with a sleeper adversary confirms the predictions: the composite-trust bounds hold across twenty thousand random draws, the analytic fixed point matches simulation to within 0.007, and a compromised prerequisite layer triggers automatic capability revocation within a few interactions.**

***Index Terms*—multi-agent systems, large language models, trust, zero-trust, capability control, agent safety, alignment**

I. Introduction

Large language model (LLM) agents are increasingly deployed not in isolation but as collaborating collectives that plan, negotiate, and act through shared tools and messages [1]–[3]. Collaboration improves capability, but it also makes *trust* between agents a first-class operational variable rather than a background assumption. A compromised or hallucinating agent whose output is consumed without scrutiny can propagate errors system-wide, and structured attacks that exploit inter-agent trust raise harmful-response rates well above single-agent baselines [24]. Recent work formalizes a *Trust-Vulnerability Paradox* (TVP): higher inter-agent trust improves task success yet monotonically increases over-exposure and authorization drift [4]. Trust is therefore better treated as a revocable, continuously verified security variable governed by least-privilege principles [5]–[7].

Layered trust models are an appealing response: they decompose trustworthiness into dimensions such as epistemic reliability, behavioral alignment, role awareness, social accountability, and institutional governance. Yet existing layered accounts are largely descriptive. Three questions are left open. (i) *Composition*: how do layer-wise trust values combine into a single actionable score, given that the layers are not independent? (ii) *Weighting*: how is the relative importance of each layer set, and can it adapt to which layer is actually failing? (iii) *Actuation*: how should the resulting trust govern what an agent is permitted to do? Without answers, a layered model cannot be deployed at runtime.

This paper closes these gaps. We treat the layered stack as a dynamical object and make it *operational*. **Contributions:**

- **A cross-layer synergy operator** (Sec. III) propagating prerequisite-layer deficits into dependent layers, with a generalized-mean composite trust Φ_p whose limit $p \to -\infty$ recovers the weakest-link rule (Theorem 1, Proposition 2), and a closed-form conservative trust fixed point (Proposition 1).
- **Failure-grounded adaptive weights** (Sec. IV): layer importances $\alpha_\ell(t)$ learned online from layer-attributed failures with a no-regret guarantee (Lemma 2).
- **Trust-Gated Capability Control** (Sec. V): a runtime algorithm mapping composite trust to revocable capability grants. Theorem 2 proves it breaks the TVP, and Corollary 1 bounds the revocation latency.
- **A numerical study** (Sec. VI) confirming the bounds, the fixed point, and self-revocation under a sleeper adversary.

We also map the framework onto deployed agent protocols (MCP, A2A) and sketch cross-modal and economic-incentive extensions (Sec. VII).

II. Related Work

Trust as a security variable. Qi et al. [4] formalize the Trust–Vulnerability Paradox, isolating three facets of inter-agent trust: its *strength* (how readily one agent accepts another's claims), *scope* (which information and actions are thereby authorized), and *revocability* (how quickly permissions can be withdrawn). They show empirically that raising a scalar trust level τ monotonically increases both an over-exposure rate and an authorization-drift metric across several agent frameworks. Their defenses (sensitive-information repartitioning and a guardian agent that polices disclosures)

reduce exposure but operate *around* the trust signal rather than reshaping how it is computed. Zero-trust treatments for agentic AI [5], [7] push in a complementary direction, recasting trust as never-implicit and continuously re-verified, with verifiable agent identity, least-privilege scopes, and per-call revocation; TRiSM [6] organizes the broader trust, risk, and security-management landscape. These works establish *that* trust must be a first-class, revocable control variable, but leave open *how* a multi-dimensional trust signal should be aggregated and turned into authorization. This is precisely the gap our compositional operator and gating rule fill.

Dynamic and probabilistic trust. Reputation systems in classical multi-agent research summarize an agent's past behavior into a scalar score [8], frequently with Beta or Dirichlet posteriors whose conjugacy makes online updates cheap [9]. Recent LLM-era work revives this idea: DynaTrust [10] maintains per-agent Beta beliefs with asymmetric, failure-weighted smoothing to expose *sleeper* agents whose behavior shifts after a dormant period, while AgentGuard [11] learns an online Markov-decision-process model of an agent's emergent behavior for runtime probabilistic assurance; economic-game probes have even been used to *elicit* the trustworthiness priors LLMs themselves hold [27]. All of these track a *single* trust quantity per agent and are therefore blind to the situation at the heart of the paradox: an agent that is behaviorally cooperative yet epistemically broken. We retain the Beta machinery, and prove its conservative steady state, but lift it to a *vector* of layer trusts coupled by a synergy operator, which is what lets a hidden prerequisite failure veto authorization.

Multi-agent debate, deception, and collective safety. Collaboration among LLM agents can both improve and degrade safety. Structured debate raises answer quality and factuality [25], yet the same channels are exploitable: structured jailbreaks injected into one debater propagate through inter-agent messages and lift harmful-response rates well above single-agent baselines [24]. Simulation studies of deception and betrayal further show that agents readily form, exploit, and misjudge trust in social settings [26]. Such findings argue for treating a compromised collaborator as the default threat rather than an edge case (the stance our threat model adopts), and for authorization that does not amplify a single corrupted agent's influence across the collective.

Uncertainty, calibration, and epistemic reliability. Any trust model's epistemic layer rests on knowing when an agent is wrong. Bayesian and ensemble methods quantify predictive uncertainty [14], but LLMs are frequently *miscalibrated*, reporting high confidence on incorrect outputs; recent work studies how to build and propagate well-grounded epistemic state across collaborating agents [28]. Our framework consumes such signals through the epistemic layer's evidence s_1 and, via the synergy operator, lets a calibration collapse cascade into reduced authorization, turning a passive reliability measurement into an active control input.

Agent protocols. Interoperability standards such as MCP [16], A2A [17], and the broader protocol landscape [18] expose agent capabilities as scoped, token-mediated grants, and increasingly issue short-lived credentials in the spirit of least privilege. They standardize the *mechanism* of capability exchange but are deliberately silent on the *policy* that decides which scopes to issue. TGCC supplies exactly this missing policy layer (Sec. VII). Table I positions our approach against these lines of work: prior methods are dynamic, or revocable, or partially layered, but none combines cross-layer composition, failure-adaptive weighting, capability gating, and formal guarantees.

TABLE I
POSITIONING AGAINST PRIOR TRUST AND SAFETY APPROACHES. *Comp.*: COMPOSITIONAL CROSS-LAYER TRUST; *Adapt.*: FAILURE-ADAPTIVE LAYER WEIGHTS; *Gate*: CAPABILITY GATING/REVOCATION; *Guar.*: PROVEN GUARANTEES.

Approach	Comp.	Adapt.	Gate	Guar.
Reputation [8], [9]	×	×	×	×
TVP defenses [4]	×	×	partial	×
DynaTrust [10]	×	×	×	×
Zero-trust [5], [7]	×	×	✓	×
TGCC (ours)	✓	✓	✓	✓

III. The Operational Trust Stack

> ***Core idea.*** *Maintain a* vector *of per-layer trusts, couple them so a prerequisite-layer deficit suppresses the layers that depend on it, and gate every capability on the resulting composite trust. Authorization then tracks the weakest* verified *layer, not the raw trust an adversary can inflate.*

A. Threat model and assumptions

Setting. N agents collaborate, each consuming others' outputs and requesting capabilities (tool calls, write actions, or message scopes) mediated by a controller that decides grants. **Adversary.** A compromised *sleeper* agent first behaves cooperatively to accrue trust and, once granted high-risk capabilities, acts maliciously. Critically, it can keep its *behavioral* conduct agreeable while a prerequisite layer degrades, typically its *epistemic* reliability collapsing into confident hallucination. **Assumptions.** (A1) each layer ℓ exposes a check yielding a noisy but informative success signal s_ℓ (calibration probe, role-conformance test, audit trail, etc.); (A2) the layer dependency structure is known and encoded by a fixed lower-triangular coupling matrix $\mathbf{C}$, with epistemic reliability a prerequisite of all higher layers; (A3) per-capability risk thresholds are set by the deployer. **Goal.** Prevent privilege escalation by a stealthily compromised agent while preserving capability for honest agents: authorization should follow verified cross-layer trust, not raw behavioral agreement.

B. Per-layer trust as a Beta belief

Consider $L = 5$ ordered trust layers indexed $\ell \in \{1, \dots, L\}$: epistemic (1), behavioral (2), role (3), social (4), institutional (5). For agent i at step t we maintain a trust vector $\mathbf{T}_i(t) = [T_{i,1}(t), \dots, T_{i,L}(t)]^\top \in [0,1]^L$ (we drop the index i where a single agent is meant). Each component is the mean of a Beta

posterior accumulating per-layer evidence with a forgetting factor $\gamma \in (0,1]$ and an asymmetric penalty $\omega \geq 1$ on failures:

$$a_\ell \leftarrow \gamma a_\ell + \mathbb{1}[s_\ell], \qquad b_\ell \leftarrow \gamma b_\ell + \omega \mathbb{1}[\neg s_\ell], \tag{1}$$

$$T_{i,\ell}(t) = \frac{a_\ell}{a_\ell + b_\ell}, \tag{2}$$

where s_ℓ is the success indicator of a layer-ℓ check (calibration probe, utility-divergence test, role-conformance check, audit, etc.). The forgetting factor keeps trust responsive to behavioral shifts; $\omega > 1$ makes trust fall faster than it rises, as is standard for safety-critical reputation [9], [10]. The estimator admits a closed-form, provably conservative steady state.

Proposition 1 (Conservative trust fixed point)**.** *Under stationary layer reliability* $\Pr[s_\ell] = \rho_\ell$*, the expected-evidence recursion induced by* (1) *with forgetting* $\gamma \in (0,1)$ *and penalty* $\omega \geq 1$ *has the unique fixed point*

$$T_\ell^\star = \frac{\rho_\ell}{\rho_\ell + \omega(1-\rho_\ell)}, \tag{3}$$

which satisfies $T_\ell^\star \leq \rho_\ell$ *with equality iff* $\omega = 1$*, is strictly increasing in* ρ_ℓ*, and is reached geometrically at rate* γ *(independent of its value on* γ*).*

Proof. Taking expectations of (1) gives two linear recursions, contractive since $\gamma < 1$:

$$\bar{a}_{t+1} = \gamma \bar{a}_t + \rho_\ell, \qquad \bar{b}_{t+1} = \gamma \bar{b}_t + \omega(1-\rho_\ell),$$

with fixed points $\bar{a}^\star = \rho_\ell/(1-\gamma)$ and $\bar{b}^\star = \omega(1-\rho_\ell)/(1-\gamma)$. Hence

$$T_\ell^\star = \frac{\bar{a}^\star}{\bar{a}^\star + \bar{b}^\star} = \frac{\rho_\ell}{\rho_\ell + \omega(1-\rho_\ell)},$$

the factor $1-\gamma$ cancelling. Strict monotonicity follows from $\partial T_\ell^\star / \partial \rho_\ell = \omega\,[\rho_\ell + \omega(1-\rho_\ell)]^{-2} > 0$, and

$$T_\ell^\star \leq \rho_\ell \iff (\omega - 1)(1-\rho_\ell) \geq 0,$$

which holds for $\omega \geq 1$ (equality iff $\omega = 1$). Contractivity at rate γ gives the stated geometric convergence. □

For example, $(\rho, \omega) = (0.9, 3)$ gives $T^\star = 0.75$ and $\rho = 0.3$ gives $T^\star \approx 0.13$, so a reliability collapse is amplified into a sharp, conservative trust drop, the behavior exploited by the controller in Sec. V.

C. Cross-layer synergy operator

The layers are not independent: a deficit in a prerequisite layer should suppress the usable trust of every layer that depends on it. We encode dependencies by a lower-triangular *coupling matrix* $\mathbf{C} \in [0,1]^{L \times L}$, where $C_{\ell k}$ (for $k < \ell$) is the degree to which layer ℓ is conditioned on prerequisite k.

Definition 1 (Synergy operator)**.** The *effective trust* of layer ℓ is

$$\tilde{T}_\ell \;=\; T_\ell \prod_{k<\ell} \big(1 - C_{\ell k}(1 - T_k)\big). \tag{4}$$

A prerequisite deficit $(1 - T_k)$ discounts $\tilde{T}_\ell$ by a factor that scales with the coupling $C_{\ell k}$: when $C_{\ell k} = 0$ there is no dependency, and when $C_{\ell k} = 1$ a fully failed prerequisite ($T_k = 0$) zeroes the contribution. Equation (4) is a precise instantiation of the otherwise informal notion that "failure at a lower layer propagates upward."

D. Composite trust

We aggregate the effective trusts into a single score via a weighted generalized (power) mean of order $p < 0$ with adaptive weights $\alpha_\ell \geq 0$, $\sum_\ell \alpha_\ell = 1$:

$$\Phi_p(\mathbf{T}; \boldsymbol{\alpha}) \;=\; \Big(\textstyle\sum_{\ell=1}^{L} \alpha_\ell \tilde{T}_\ell^{\,p}\Big)^{1/p}. \tag{5}$$

Negative p makes Φ_p sensitive to the weakest layers; $p \to -\infty$ recovers the conservative weakest-link rule $\min_\ell \tilde{T}_\ell$, while moderate $|p|$ permits limited compensation between healthy layers. The next results make this precise. We first verify the construction is well-posed.

Lemma 1 (Well-posedness)**.** *For any* $\mathbf{T} \in [0,1]^L$ *and lower-triangular* $\mathbf{C} \in [0,1]^{L \times L}$*, the effective trusts satisfy* $\tilde{T}_\ell \in [0,1]$*, and for any weights* $\boldsymbol{\alpha}$ *on the simplex and* $p < 0$*,* $\Phi_p \in [\min_\ell \tilde{T}_\ell,\ \max_\ell \tilde{T}_\ell] \subseteq [0,1]$*. Hence the stack is a well-defined map* $[0,1]^L \to [0,1]$*.*

Proof. Each factor $1 - C_{\ell k}(1 - T_k) \in [0,1]$ because $C_{\ell k}, T_k \in [0,1]$; a finite product of $[0,1]$ terms times $T_\ell \in [0,1]$ stays in $[0,1]$, so $\tilde{T}_\ell \in [0,1]$. A weighted power mean of arguments in $[0,1]$ lies between their minimum and maximum, both in $[0,1]$. □

Theorem 1 (Bounds and weakest-link limit)**.** *For any* $p < 0$*,* Φ_p *is non-decreasing in each* $\tilde{T}_\ell$ *and satisfies*

$$\min_\ell \tilde{T}_\ell \;\leq\; \Phi_p(\mathbf{T}; \boldsymbol{\alpha}) \;\leq\; \sum_\ell \alpha_\ell \tilde{T}_\ell, \tag{6}$$

with $\lim_{p \to -\infty} \Phi_p = \min_\ell \tilde{T}_\ell$*. Moreover,*

$$\Phi_p(\mathbf{T}; \boldsymbol{\alpha}) \;\leq\; \kappa \min_\ell \tilde{T}_\ell, \qquad \kappa := \big(\min_\ell \alpha_\ell\big)^{1/p} \geq 1. \tag{7}$$

Proof. Bounds (6) and the limit are the standard ordering of weighted power means: the mean of order p is non-decreasing in p, equals the arithmetic mean at $p = 1$ and the minimum at $p \to -\infty$; monotonicity of Φ_p in each $\tilde{T}_\ell$ follows since each summand $\tilde{T}_\ell^{\,p}$ is monotone. For (7), let $j \in \arg\min_\ell \tilde{T}_\ell$. As $p < 0$ every term is positive, so

$$\sum_\ell \alpha_\ell \tilde{T}_\ell^{\,p} \;\geq\; \alpha_j\, \tilde{T}_j^{\,p}.$$

The map $x \mapsto x^{1/p}$ is decreasing for $p < 0$; applying it and using $\alpha_j \geq \min_\ell \alpha_\ell$,

$$\Phi_p \;\leq\; \alpha_j^{1/p}\, \tilde{T}_j \;\leq\; \big(\min_\ell \alpha_\ell\big)^{1/p} \min_\ell \tilde{T}_\ell \;=\; \kappa \min_\ell \tilde{T}_\ell. \quad \square$$

Equation (7) is the key actuation lever: composite trust is pinned to the weakest effective layer up to a bounded multiplicative slack κ. We next bound how a single compromised prerequisite caps every dependent layer.

Proposition 2 (Cascade containment)**.** *Suppose a prerequisite layer* k *is compromised,* $T_k \leq \delta$*. Then every dependent layer*

$\ell > k$ with $C_{\ell k} \geq c$ obeys $\tilde{T}_\ell \leq 1 - c(1-\delta)$. In particular, if the epistemic layer ($k = 1$) is compromised and $C_{\ell 1} \geq c$ for all ℓ, then $\Phi_p \leq \kappa\,\delta$.

Proof. In (4) the factor for prerequisite k is $1 - C_{\ell k}(1 - T_k) \leq 1 - c(1-\delta)$; all other factors and T_ℓ are ≤ 1, giving the per-layer bound. If $k = 1$ then $\tilde{T}_1 = T_1 \leq \delta$, so $\min_\ell \tilde{T}_\ell \leq \delta$ and (7) gives $\Phi_p \leq \kappa\delta$. □

IV. Failure-Grounded Adaptive Weighting

The weights $\boldsymbol{\alpha}(t)$ decide which layers the composite score should attend to. We argue they should track *which layer is currently most responsible for harm*, information available from runtime failure attribution. Let $q_\ell(t) \in [0,1]$ be a layer-attributed loss (e.g., the fraction of harmful episodes in a window whose root cause is layer ℓ). We maintain an exponentially weighted risk estimate and a multiplicative-weights (Hedge) update:

$$\hat{r}_\ell(t) = (1-\lambda)\hat{r}_\ell(t-1) + \lambda\, q_\ell(t), \tag{8}$$

$$\alpha_\ell(t) = \frac{\exp\big(\eta\,\hat{r}_\ell(t)\big)}{\sum_k \exp\big(\eta\,\hat{r}_k(t)\big)}. \tag{9}$$

This places weight on layers whose failures most predict system harm, so the composite trust becomes most sensitive precisely where the system is breaking. Treating each layer as an "expert" whose instantaneous loss is $-\hat{r}_\ell(t)$, (9) is the Hedge algorithm and inherits its guarantee.

Lemma 2 (No-regret layer tracking)**.** *With learning rate $\eta = \sqrt{8 \ln L / T}$, the weighting (9) achieves regret $R_T \leq \sqrt{(T/2)\ln L}$ relative to the best fixed layer-weighting in hindsight over a horizon T.*

Proof. Immediate from the standard regret bound of the Hedge/multiplicative-weights algorithm with bounded losses in $[0,1]$ [19], [20]. □

Thus $\boldsymbol{\alpha}(t)$ converges, in the no-regret sense, to emphasizing the empirically dominant failure layer, linking the system's failure record directly to its trust computation rather than fixing weights a priori.

V. Trust-Gated Capability Control

We now actuate composite trust. Let $\mathcal{C}$ be the set of capabilities (tool calls, message scopes, write actions). Each $c \in \mathcal{C}$ carries a risk threshold $\theta_c \in [0,1]$ and a set of *required prerequisite layers* $\mathrm{req}(c)$ with per-layer thresholds $\theta_{c,\ell}$.

Definition 2 (Trust-gated grant)**.** Agent i is granted capability c at step t iff

$$g_i(c,t) = \mathbb{1}\Big[\Phi_p(\mathbf{T}_i(t)) \geq \theta_c \;\wedge\; \forall \ell \in \mathrm{req}(c)\colon\; \tilde{T}_{i,\ell}(t) \geq \theta_{c,\ell}\Big]. \tag{10}$$

Grants are recomputed every step, so a capability token is short-lived and automatically revoked when trust degrades, realizing least-privilege and revocability. Algorithm 1 gives the runtime loop.

Algorithm 1 TGCC runtime loop (per agent i, per step t)

Require: coupling $\mathbf{C}$, order p, thresholds $\{\theta_c, \theta_{c,\ell}\}$, rates λ, η

observe layer checks $s_1, \ldots, s_L$; update Beta beliefs via (1)
compute effective trust $\tilde{\mathbf{T}}$ via synergy operator (4)
update risk $\hat{r}_\ell$ and weights $\boldsymbol{\alpha}$ via (9)
composite $\Phi_p \leftarrow (\sum_\ell \alpha_\ell \tilde{T}_\ell^{\,p})^{1/p}$
for all capabilities $c \in \mathcal{C}$ **do**
    issue/keep token for c iff grant predicate (10) holds; else revoke
end for
if $\min_\ell \tilde{T}_\ell < \theta_{\mathrm{rec}}$ **then**
    trigger layer-specific recovery (e.g. external knowledge anchoring)
end if

Because every quantity is maintained by a recursive update, the controller is cheap enough to run on every interaction.

Proposition 3 (Constant per-step overhead)**.** *Each iteration of Algorithm 1 runs in $O(L^2 + |\mathcal{C}|)$ time and $O(L)$ memory per agent, independent of the interaction history length t.*

Proof. The Beta updates (1) and the Hedge update (9) touch each layer once ($O(L)$) and store only $2L$ pseudo-counts and L risk estimates. The synergy operator (4) forms L products of at most L factors ($O(L^2)$), and the composite (5) is $O(L)$. The grant test (10) costs $O(|\mathcal{C}|)$ for bounded prerequisite sets. History-independence follows because the forgetting recursions summarize the entire past in the $O(L)$ state. □

With $L = 5$ this overhead is negligible, so per-step re-evaluation (and hence prompt revocation) is practical even in large agent collectives.

A. Breaking the Trust-Vulnerability Paradox

A naive controller authorizes on a single pairwise/behavioral trust signal $\tau \approx T_2$ (the behavioral layer). Under such gating, an adversary that keeps behaving agreeably retains privilege even while its epistemic reliability collapses. This is precisely the regime in which over-exposure grows with trust [4]. TGCC removes this attack surface.

Theorem 2 (Paradox resolution)**.** *Consider a stealthy compromise in which an agent's behavioral trust is driven arbitrarily high, $T_2 \to 1$, while a prerequisite layer k degrades to $T_k \leq \delta$. Under naive behavioral gating, every capability with $\theta_c \leq T_2$ remains granted, so over-exposure is unbounded in the compromise depth. Under TGCC with order p and k a prerequisite of all layers (e.g. epistemic), every capability with*

$$\theta_c \;>\; \kappa\,\delta, \qquad \kappa = (\min_\ell \alpha_\ell)^{1/p}, \tag{11}$$

is revoked irrespective of T_2. Hence the granted high-risk set is non-increasing in the compromise depth $(1-\delta)$.

Proof. By Proposition 2 the compromise gives $\Phi_p \leq \kappa\delta$. If $\theta_c > \kappa\delta$ then $\Phi_p < \theta_c$, so the first conjunct of (10) fails

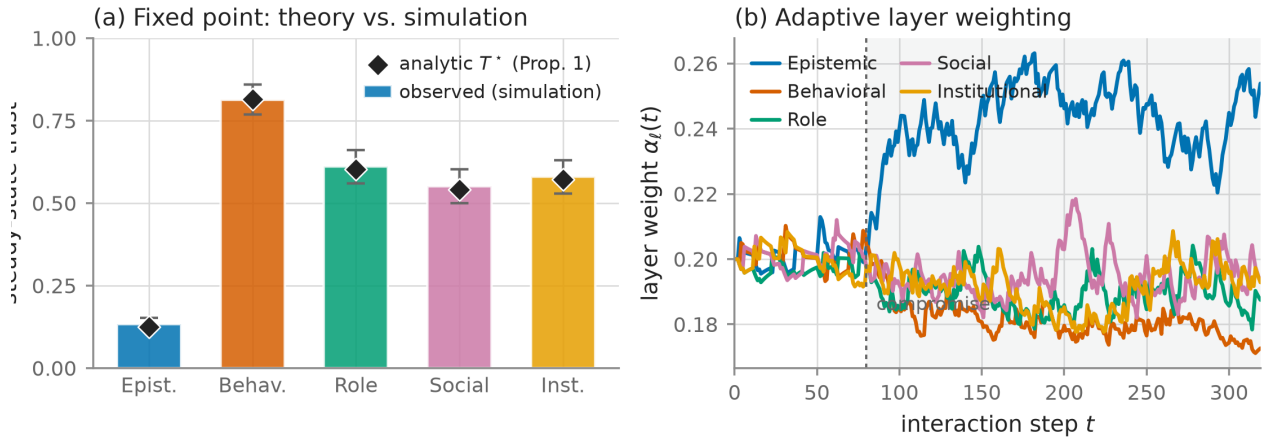


Fig. 1. (a) Proposition 1 validation: observed steady-state trust (bars) vs. the analytic fixed point $T^\star$ (diamonds), mean absolute error 0.007. (b) Failure-grounded weights $\alpha_\ell(t)$ shift toward the epistemic layer after compromise (Lemma 2).

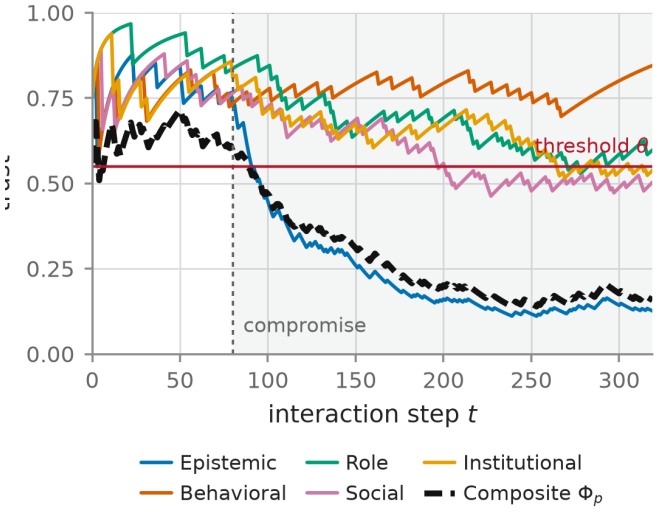


Fig. 2. Trust dynamics of a sleeper agent. Behavioral trust stays high; epistemic collapse propagates through the synergy operator and pushes composite trust Φ_p below the revocation threshold θ (self-revocation, Corollary 1).

and c is denied regardless of T_2. Decreasing δ (a deeper compromise) lowers the bound $\kappa\delta$, weakly enlarging the revoked set, so the granted high-risk set is non-increasing in $(1-\delta)$. The naive case is immediate, its grant rule depending only on T_2. □

Combining the conservative fixed point (Proposition 1) with Theorem 2 bounds how quickly a compromise is contained.

Corollary 1 (Self-revocation with bounded latency)**.** *Suppose at time t_0 a prerequisite layer k (prerequisite to all layers) suffers a reliability drop whose conservative fixed point satisfies $T_k^\star \le \delta$. Then $T_k(t) \to T_k^\star$ geometrically at rate γ, and once $T_k(t) \le \delta$, TGCC denies every capability with $\theta_c > \kappa\delta$ regardless of behavioral trust. The number of steps until T_k reaches an ϵ-neighborhood of $T_k^\star$ is $O\big(\log_{1/\gamma}(1/\epsilon)\big)$.*

Proof. Geometric convergence at rate γ follows from Proposition 1; the threshold-crossing time is then logarithmic in the target accuracy. Once $T_k \le \delta$, Theorem 2 applies. □

> ***Key insight.*** *Naive gating makes authorization* increasing *in raw trust; TGCC makes it increasing in the* weakest cross-layer-verified *trust, which an adversary cannot inflate by behaving agreeably. The paradox's monotone exposure growth becomes bounded, self-revoking exposure with logarithmic detection latency.*

VI. Numerical Illustration

We validate the framework in a controlled simulation of the layer-trust dynamics, instantiating a single agent under a *sleeper*-agent threat of the kind arising in long-running collectives such as Agent Hospital [22], MetaGPT [21], and GPTSwarm [23]. Each layer emits Bernoulli evidence from a latent reliability ρ_ℓ; beliefs update via (1) ($\gamma = 0.985$, $\omega = 3$). Before activation at $t = 80$ all layers are reliable ($\rho \approx 0.9$); after activation the agent's epistemic reliability collapses to $\rho_1 = 0.30$ (confident hallucination) while its behavioral reliability stays at 0.93 (it remains agreeable), matching the exact stealth profile of Theorem 2. We set epistemic as a prerequisite of all layers, $p = -6$, and revocation threshold $\theta = 0.55$. Statistics are averaged over 400 seeds.

Mechanism (Fig. 2). Behavioral trust stays high after activation, yet epistemic trust collapses and, through the synergy operator (4), drags the composite Φ_p below θ, so TGCC revokes the high-risk capability while a behavioral-only gate never would. This is Theorem 2 made visible.

Validation of the theory. Three checks (Table II). (i) *Theorem 1*: across 20,000 random draws of $(\mathbf{T}, \boldsymbol{\alpha})$ we found zero violations of the bounds (6). (ii) *Proposition 1*: the observed steady-state trust matches the closed-form fixed point $T_\ell^\star$ to a mean absolute error of 0.007 across layers (Fig. 1(a)). (iii) *Corollary 1*: after compromise the capability is revoked within 3.1 steps on average (median 2), consistent with the predicted logarithmic latency. Fig. 1(b) shows the Hedge

TABLE II
PREDICTIONS VS. SIMULATION (400 SEEDS).

Result	Prediction	Observed
Thm. 1 bound violations	0	0/20,000
Prop. 1 fixed point (MAE)	exact	0.007
Cor. 1 revocation latency	$O(\log_{1/\gamma}\frac{1}{\epsilon})$	3.1 steps
Breaks TVP (Thm. 2)	yes	yes

weights shifting toward the epistemic layer once it becomes the dominant failure source, as Lemma 2 anticipates.

VII. DEPLOYMENT AND EXTENSIONS

Protocol mapping. TGCC is directly implementable over emerging agent protocols. In MCP [16] and A2A [17], capabilities are exposed as scoped, token-mediated grants but the policy that sets scope is unspecified [18]. Equation (10) is exactly such a policy: a capability's OAuth-style scope is issued only while Φ_p and the required prerequisite layers clear the capability's thresholds, and short-lived tokens give the per-call revocation that zero-trust guidance recommends [5], [7].

Cross-modal epistemic trust. The framework is modality-agnostic: for embodied or perceptual agents, the epistemic layer's evidence s_1 can be supplied by sensor-fusion confidence or perceptual anomaly scores, so that adversarial perceptual perturbations degrade T_1 and, via (4), gate downstream actuation, extending the stack beyond dialogue agents.

Economic-incentive layer. In agentic marketplaces, misaligned incentives (e.g. tacit collusion) are not captured by reliability alone. An incentive layer between role and social, whose evidence tracks deviation from incentive-compatible behavior, slots into (4) without changing the operator, and composes with governance-graph mechanisms [15].

VIII. CONCLUSION

We turned a descriptive layered trust model into an operational controller for multi-agent LLM systems. A cross-layer synergy operator and a generalized-mean composite give a principled, bounded way to combine layer trusts; a no-regret estimator grounds layer importance in observed failures; and Trust-Gated Capability Control converts the result into revocable, least-privilege authorization. We proved, and numerically confirmed, that this combination breaks the Trust-Vulnerability Paradox: privilege tracks the weakest verified layer, not raw trust, so a stealthy compromise that inflates behavioral trust while degrading a prerequisite layer self-revokes within a logarithmic number of steps rather than escalating. The framework is implementable over current agent protocols and extends to perceptual and economic settings, offering a concrete path to safe, scalable multi-agent collaboration.

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